



INFORMATION TECHNOLOGY: PAPER II

Time: 3 hours

120 marks

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This paper consists of 7 pages. Please check that your question paper is complete.
2. This question paper is to be answered using Object Oriented Programming principles. Your programme must make sensible use of methods and parameters.
3. This paper is divided into two questions. All candidates must answer both questions.
4. This paper is written in programming terms that are not specific to any particular programming language. There is no distinction made with regards to the programming language that you are using. Where code has been provided for you, specific instructions have been provided for Java and Delphi candidates respectively.
5. Make sure that you answer the questions in the manner described because marks will be awarded for your solution according to the specifications that are given in the question.
6. Only answer the question that is stated. For example, if the question does not ask for data validation, then no marks are awarded for it, and therefore no code needs to be written.
7. If you cannot get a section of code to work, comment it out so that it will not be executed and so that you can continue with the examination. If possible, try to explain the error to aid the marker.
8. When accessing a file, DO NOT use full path names of the file, as this will create problems when the programme is marked on a computer other than the one you are writing on. Merely refer to the files using their names and extensions, where necessary.
9. Your programmes must be coded in such a way that they will work with any data and not just the sample data supplied, or any data extracts that appear in the question paper. You are advised to look at the supplied data files carefully.
10. Make sure that routines such as searches, sorts and selections are developed from first principles, and that you do not use the built-in features of a programming language for any of the above. You may use SQL to sort the data if you are retrieving it from a database.
11. In visual languages, ensure that you do not use the interface or built-in components as your data structure. The data structure must be defined and declared by you, the programmer.

12. Read the whole question before you choose a data structure. You will find that there may be an alternative method of representing the data which will be more efficient considering the questions that are asked in the paper.
 13. You must save all your work regularly on the disk you have been given, or the disk space allocated to you for this examination.
 14. If there is a technical interruption that prevents you from writing your examination, such as a power failure, when you resume writing your examinations, you will only be given the time that was remaining when the interruption began. No extra time will be given to catch up with work that was not saved.
 15. Make sure that your examination number appears as a comment in every programme that you code as well as on every page of hard copy that you hand in.
 16. Print a code listing of all the programmes/classes that you code. Print the output from your programme, if possible. Printing must be done after the examination. You will be given half an hour to print after the examination is finished. Your teacher will tell you what arrangements have been made for the printing of your work.
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QUESTION 1 DATABASES AND SQL

The files required for this question can be found in the folder called **Question1**. Please complete your work in this folder or in a project called **Question1**.

You are strongly advised to make a backup copy of the files should you wish to revert back to the originals, for any reason.

Find-A-Pet, a company that implants RFID¹ microchips into pets so that they may be safely returned to their owners if ever they are lost, has kept their data in a spreadsheet until recently when they were advised that it would be much quicker to keep this data in a database.

A description of each of the fields in the tables follows:

tblOwners

OwnerID	This is a unique number associated with each owner.
LastName	This field contains the owner's last name.
FirstName	This field contains the owner's first name.
Telephone	This field contains the owner's telephone number so that he or she can be contacted if one of their pets is found.
City	This field contains the city that the owner lives in.

tblPets

PetID	This is a unique number associated with each pet. This is the number that is recorded in the RFID microchip.
PetName	This field contains the pet's name.
OwnerID	This field is a foreign key which associates the data with the tblOwners table.
PetType	This field contains the type of animal. Currently <i>Find-A-Pet</i> only stores records for pets that are either 'dog', 'cat' or 'bird'.
PetBreed	This field contains the breed of the animal. For example 'Doberman', 'Labrador', 'Persian', and 'African Grey' are just some of the breeds that are in the table.

You are provided with a single programme called **FindAPet**. This class contains the necessary code to connect to the database.

The marks for this programme are awarded for creating the correct SQL statements and not for the overall structure of the programme.

¹ RFID stands for Radio Frequency Identification. These tiny chips are injected under the animal's skin and cause them no discomfort. The chips can be detected by passing an RFID scanner over the surface of the animal. This scanner is able to read the identification number that is stored on the chip without causing any undue stress to the animal.

Open the programme named **FindAPet**.

- 1.1 **Menu Option A:** Display a list of all the owners stored in the database, sorted by their last names and then their first names. (5)
- 1.2 **Menu Option B:** Ask the user to input a city and then display all the owners who reside in that city. (5)
- 1.3 **Menu Option C:** Display how many of each type of pet (dogs, cats and birds) is stored in the database. (7)
- 1.4 **Menu Option D:** Ask the user to input a type of pet ('dog', 'cat' or 'bird') and then display all the owners who own those types of pets. (8)
- 1.5 **Menu Option E:** Write code to add a new pet to the database. The user should be prompted to type in the last name of the owner. If there is more than one owner with that last name, then all the owners with that last name need to be displayed so that the user can type in the OwnerID of the correct owner. If the owner's last name is not found in the database, display a message to the user and do not insert anything into the database. (15)

40 marks

QUESTION 2 OBJECT ORIENTED PROGRAMMING

*The files required for this question are provided in the folder called **Question2**. Please complete your work in this folder or in a new project called **Question2**.*

*You are strongly advised to create a backup copy of the three files **Delegate**, **ConferenceApp** and **conference.txt** in case you wish to revert back to the original files.*

ConferZ is a company that hosts and organises conferences for other companies. You have been asked to create a programme that will help them to manage the delegates that are coming to a conference they have organised.

Each delegate will choose various seminars to attend. These seminars are on the following programming topics:

- P – Programming
- S – SQL Queries
- O – Object Oriented Programming
- D – Database Design

A delegate may choose to attend as few as one seminar, or as many as four seminars. A delegate will then receive a seminar code which will indicate the seminars that he or she will attend.

- A delegate who has the seminar code **DS** will be attending the Database Design seminar and the SQL Queries seminar.
- A delegate who has the seminar code **SO** will be attending the SQL Queries seminar and the Object Oriented Programming seminar.

The codes for each seminar can appear in any order.

Delegates may also elect to become 'gold delegates' which entitles them to spend a night in the hotel. Gold delegates will therefore require a room number.

ConferZ is provided with a text file that contains all the data of the delegates who are attending the conference. The text file has been named **conference.txt**. The file contains many lines, but only the first five lines are shown below for your reference:

```
Linda#Ngubane#Systro#P0  
Patrick#North#Tcsi Technology#OP  
Mervin#Shaw#Gratton#SDOP#912  
Clara#Lanuschagne#Laticrond#PSD0  
Lisa#Mbebe#Systro#OPDS#822
```

Delphi coders: Start by opening the project *ConventionApp.DRP*

- 2.1 Open the file named **Delegate (Java)/uDelegate (Delphi)**. This file contains a class called **Delegate**. The properties and constructor method are provided for you.

- 2.1.1 Code a **toString** method that will return the delegate's last name, the initial of their first name and the company that the delegate works for.

For example, if the delegate's name is 'John Smith' and he works for 'Acme Inc.', the method should return 'Smith, J./Acme Inc.' (5)

- 2.1.2 Code a typed and appropriately named accessor method for the company instance variable. (3)

- 2.2 Code a class called **GoldDelegate** in a new, appropriately named file/unit. This class inherits properties and methods from the **Delegate** class. (2)

- 2.2.1 Create a property that can be used to store the room number that the delegate will be staying in. Ensure that this variable is not directly accessible from outside this class. (3)

- 2.2.2 Code an appropriate constructor method for the **GoldDelegate** class. (4)

- 2.2.3 Code an appropriately typed and appropriately named method to return the room number that the delegate is staying in. (4)

- 2.2.4 Code a **toString** method which will override the **toString** method created in the **Delegate** class. It should keep the same information, but add on the room number that the delegate will be staying in.

For example, if John Smith from Acme Inc. is staying in room 602, the method should return 'Smith, J./Acme Inc./Room 602'. (3)

- 2.3 Code a new class unit called **ConventionManager** in its own, appropriately named file/unit. (2)
- 2.3.1 Code two instance variables for this class:
- (a) an array that can store 100 of either **Delegate** or **GoldDelegate** objects; and
 - (b) a variable to keep track of the number of elements in the array. (5)
- 2.3.2 In the **ConventionManager**, code a constructor method that will read the data provided in the text file and use it to create objects in the array. Each row of the file has either four or five fields, depending on whether it represents a normal delegate or a gold delegate. Both kinds of delegate will have their first name, last name, company, and seminar code as fields, but the gold delegates will have an extra field for their room number.
- If you cannot read data in from the file, you may write specific code to add some of the delegates to the array. This will allow you to test your programme later on. However, you will not be awarded marks for this approach.* (13)
- 2.3.3 Code a method named **getDelegateList**. This method must return the details of all the delegates that are attending the conference, including the Gold Delegates. The method should return a String containing the output formatted according to the **toString** method with each delegate's details appearing on its own line. (6)
- 2.3.4 Code a method named **getGoldDelegateList**. This method should work similarly to the **getDelegateList**, but only the details of the Gold Delegates must be returned. (4)
- 2.3.5 Code a method named **sortDelegateCompany**. This method must sort the array of conference delegates based on the company that they work for. (8)
- 2.4 Java coders open the class **ConventionApp** that is provided for you. Delphi coders work in your main form. Java coders using Netbeans open a project called **Question 2** and then open the class **ConventionApp** that has been provided for you. *If you wish to create your own interface, you may do so, but no marks are awarded for doing so.*
- You are required to fill in the necessary code for each menu option. You do not need to make any adjustments to the rest of the code in this class.
- 2.4.1 Instantiate a **ConventionManager** object. (2)
- 2.4.2 **Menu Option A:** Display all the delegates attending the conference, using the appropriate method in the **ConventionManager**. (2)
- 2.4.3 **Menu Option B:** Display all the Gold Delegates attending the conference, using the appropriate method in the **ConventionManager**. (2)
- 2.4.4 **Menu Option C:** Call the appropriate method in the **ConventionManager** object to sort the delegates. (1)

- 2.4.5 **Menu Option D:** The conference organisers would like to know how many people will attend each seminar. Write the necessary code to calculate these amounts. You may add methods to any of the classes you have used in this question to work a solution.

Of the 11 marks for this question, up to 5 marks will be awarded for a working solution. Up to 6 marks will be awarded for showing evidence of method creation, parameter passing, encapsulation and other object oriented programming principles.

(11)

NB. *It is important that your solution would still work if you were given different data in the **conference.txt** file.*

80 marks

Total: 120 marks